The University of the State of New York

REGENTS HIGH SCHOOL EXAMINATION

GEOMETRY (Common Core)

Friday, June 17, 2016 — 1:15 to 4:15 p.m., only

Student Name:____

School Name: ____

The possession or use of any communications device is strictly prohibited when taking this examination. If you have or use any communications device, no matter how briefly, your examination will be invalidated and no score will be calculated for you.

Print your name and the name of your school on the lines above.

A separate answer sheet for Part I has been provided to you. Follow the instructions from the proctor for completing the student information on your answer sheet.

This examination has four parts, with a total of 36 questions. You must answer all questions in this examination. Write your answers to the Part I multiple-choice questions on the separate answer sheet. Write your answers to the questions in Parts II, III, and IV directly in this booklet. All work should be written in pen, except for graphs and drawings, which should be done in pencil. Clearly indicate the necessary steps, including appropriate formula substitutions, diagrams, graphs, charts, etc. Utilize the information provided for each question to determine your answer. Note that diagrams are not necessarily drawn to scale.

The formulas that you may need to answer some questions in this examination are found at the end of the examination. This sheet is perforated so you may remove it from this booklet.

Scrap paper is not permitted for any part of this examination, but you may use the blank spaces in this booklet as scrap paper. A perforated sheet of scrap graph paper is provided at the end of this booklet for any question for which graphing may be helpful but is not required. You may remove this sheet from this booklet. Any work done on this sheet of scrap graph paper will *not* be scored.

When you have completed the examination, you must sign the statement printed at the end of the answer sheet, indicating that you had no unlawful knowledge of the questions or answers prior to the examination and that you have neither given nor received assistance in answering any of the questions during the examination. Your answer sheet cannot be accepted if you fail to sign this declaration.

Notice…

A graphing calculator, a straightedge (ruler), and a compass must be available for you to use while taking this examination.

DO NOT OPEN THIS EXAMINATION BOOKLET UNTIL THE SIGNAL IS GIVEN.

GEOMETRY (COMMON CORE)

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High School Math Reference Sheet

1 inch = 2.54 centimeters
1 meter = 39.37 inches
1 mile = 5280 feet
1 mile = 1760 yards
1 mile = 1.609 kilometers

1 kilometer = 0.62 mile
1 pound = 16 ounces
1 pound = 0.454 kilogram
1 kilogram = 2.2 pounds
1 ton = 2000 pounds

1 cup = 8 fluid ounces
1 pint = 2 cups
1 quart = 2 pints
1 gallon = 4 quarts
1 gallon = 3.785 liters
1 liter = 0.264 gallon
1 liter = 1000 cubic centimeters

Triangle	$A = \frac{1}{2}bh$
Parallelogram	$A = bh$
Circle	$A = \pi r^2$
Circle	$C = \pi d$ or $C = 2\pi r$
General Prisms	$V = Bh$
Cylinder	$V = \pi r^2 h$
Sphere	$V = \frac{4}{3}\pi r^3$
Cone	$V = \frac{1}{3}\pi r^2 h$
Pyramid	$V = \frac{1}{3}Bh$

Pythagorean Theorem	$a^2 + b^2 = c^2$
Quadratic Formula	$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$
Arithmetic Sequence	$a_n = a_1 + (n - 1)d$
Geometric Sequence	$a_n = a_1 r^{n-1}$
Geometric Series	$S_n = \frac{a_1 - a_1 r^n}{1 - r}$ where $r \neq 1$
Radians	1 radian $= \frac{180}{\pi}$ degrees
Degrees	1 degree $= \frac{\pi}{180}$ radians
Exponential Growth/Decay	$A = A_0 e^{k(t - t_0)} + B_0$

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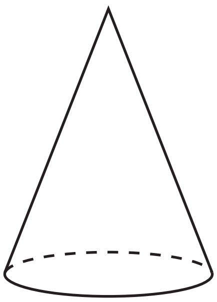
Part I

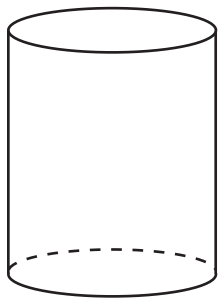
Answer all 24 questions in this part. Each correct answer will receive 2 credits. No partial credit will be allowed. Utilize the information provided for each question to determine your answer. Note that diagrams are not necessarily drawn to scale. For each statement or question, choose the word or expression that, of those given, best completes the statement or answers the question. Record your answers on your separate answer sheet. [48]

Use this space for computations.

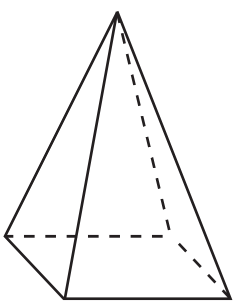
1 A student has a rectangular postcard that he folds in half lengthwise. Next, he rotates it continuously about the folded edge. Which three-dimensional object below is generated by this rotation?



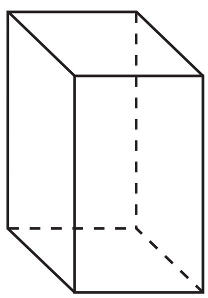
(1)



(3)



(2)



(4)

2 A three-inch line segment is dilated by a scale factor of 6 and centered at its midpoint. What is the length of its image?

(1) 9 inches
(3) 15 inches
(2) 2 inches
(4) 18 inches

Use this space for computations.

3 Kevin's work for deriving the equation of a circle is shown below.

$$x^2 + 4x = -(y^2 - 20)$$

STEP 1 $x^2 + 4x = -y^2 + 20$

STEP 2 $x^2 + 4x + 4 = -y^2 + 20 - 4$

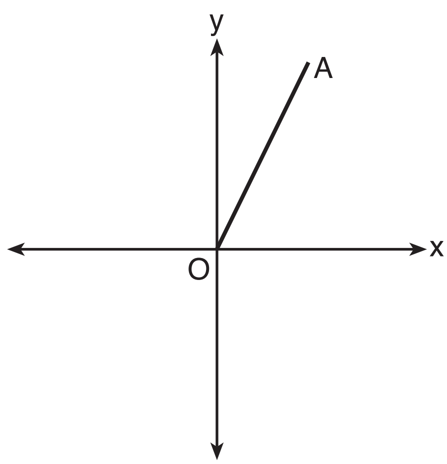
STEP 3 $(x + 2)^2 = -y^2 + 20 - 4$

STEP 4 $(x + 2)^2 + y^2 = 16$

In which step did he make an error in his work?

(1) Step 1
(2) Step 2
(3) Step 3
(4) Step 4

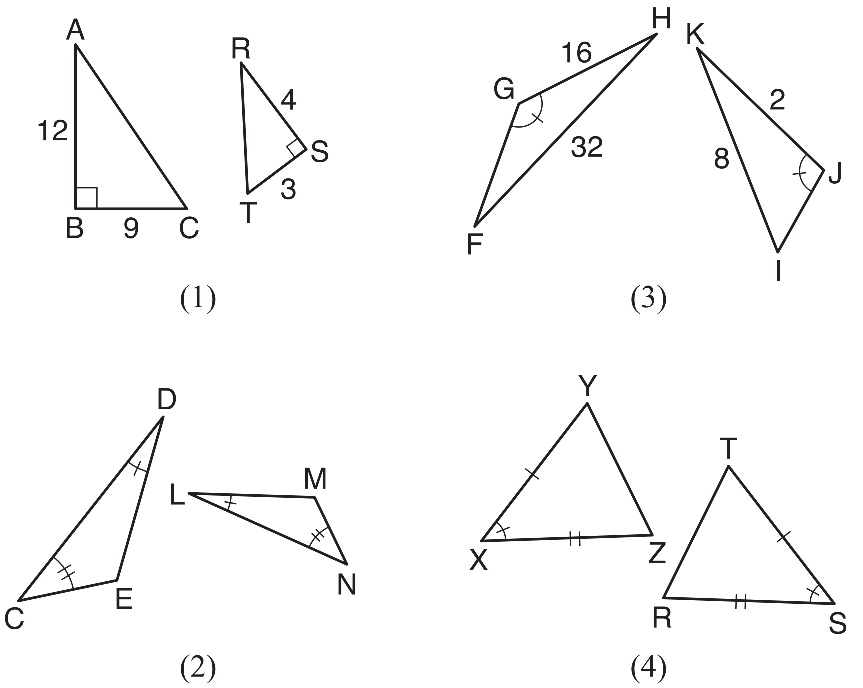
4 Which transformation of $\overline{OA}$ would result in an image parallel to $\overline{OA}$?



(1) a translation of two units down
(2) a reflection over the x-axis
(3) a reflection over the y-axis
(4) a clockwise rotation of 90° about the origin

Use this space for computations.

5 Using the information given below, which set of triangles can *not* be proven similar?



6 A company is creating an object from a wooden cube with an edge length of 8.5 cm. A right circular cone with a diameter of 8 cm and an altitude of 8 cm will be cut out of the cube. Which expression represents the volume of the remaining wood?

(1) $(8.5)^3 - \pi(8)^2(8)$

(2) $(8.5)^3 - \pi(4)^2(8)$

(3) $(8.5)^3 - \frac{1}{3}\pi(8)^2(8)$

(4) $(8.5)^3 - \frac{1}{3}\pi(4)^2(8)$

7 Two right triangles must be congruent if

(1) an acute angle in each triangle is congruent

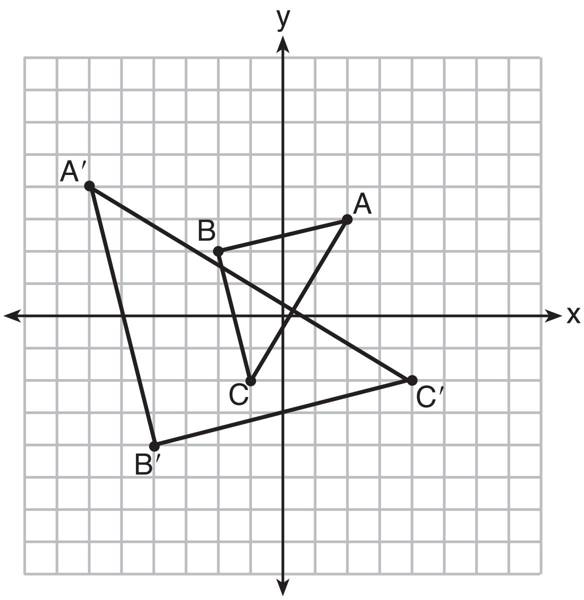
(2) the lengths of the hypotenuses are equal

(3) the corresponding legs are congruent

(4) the areas are equal

Use this space for computations.

8 Which sequence of transformations will map $\triangle ABC$ onto $\triangle A'B'C'$?



(1) reflection and translation

(2) rotation and reflection

(3) translation and dilation

(4) dilation and rotation

9 In parallelogram $ABCD$, diagonals $\overline{AC}$ and $\overline{BD}$ intersect at E. Which statement does *not* prove parallelogram $ABCD$ is a rhombus?

(1) $\overline{AC} \cong \overline{DB}$

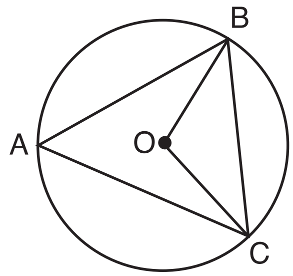
(2) $\overline{AB} \cong \overline{BC}$

(3) $\overline{AC} \perp \overline{DB}$

(4) $\overline{AC}$ bisects $\angle DCB$.

Use this space for computations.

10 In the diagram below of circle O, $\overline{OB}$ and $\overline{OC}$ are radii, and chords $\overline{AB}$, $\overline{BC}$, and $\overline{AC}$ are drawn.



Which statement must always be true?

(1) $\angle BAC \cong \angle BOC$

(2) $m\angle BAC = \frac{1}{2} m\angle BOC$

(3) $\triangle BAC$ and $\triangle BOC$ are isosceles.

(4) The area of $\triangle BAC$ is twice the area of $\triangle BOC$.

11 A 20-foot support post leans against a wall, making a 70° angle with the ground. To the *nearest tenth of a foot*, how far up the wall will the support post reach?

(1) 6.8
(2) 6.9
(3) 18.7
(4) 18.8

12 Line segment NY has endpoints $N(-11,5)$ and $Y(5,-7)$. What is the equation of the perpendicular bisector of $\overline{NY}$?

(1) $y + 1 = \frac{4}{3}(x + 3)$

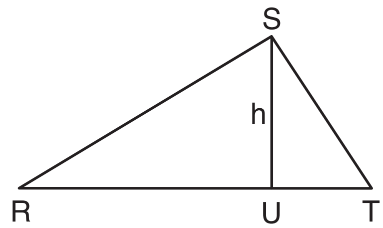
(2) $y + 1 = -\frac{3}{4}(x + 3)$

(3) $y - 6 = \frac{4}{3}(x - 8)$

(4) $y - 6 = -\frac{3}{4}(x - 8)$

Use this space for computations.

13 In $\triangle RST$ shown below, altitude $\overline{SU}$ is drawn to $\overline{RT}$ at U.



If $SU = h$, $UT = 12$, and $RT = 42$, which value of h will make $\triangle RST$ a right triangle with $\angle RST$ as a right angle?

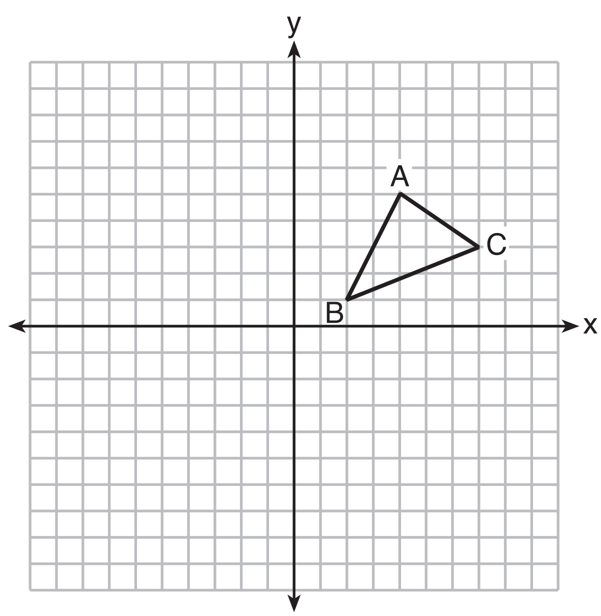
(1) $6\sqrt{3}$

(2) $6\sqrt{10}$

(3) $6\sqrt{14}$

(4) $6\sqrt{35}$

14 In the diagram below, $\triangle ABC$ has vertices $A(4,5)$, $B(2,1)$, and $C(7,3)$.



What is the slope of the altitude drawn from A to $\overline{BC}$?

(1) $\frac{2}{5}$

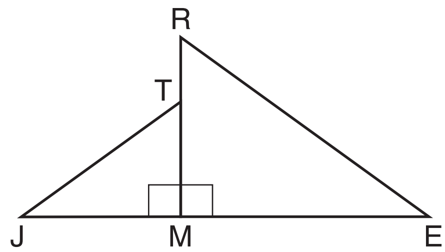
(2) $\frac{3}{2}$

(3) $-\frac{1}{2}$

(4) $-\frac{5}{2}$

Use this space for computations.

15 In the diagram below, $\triangle ERM \sim \triangle JTM$.



Which statement is always true?

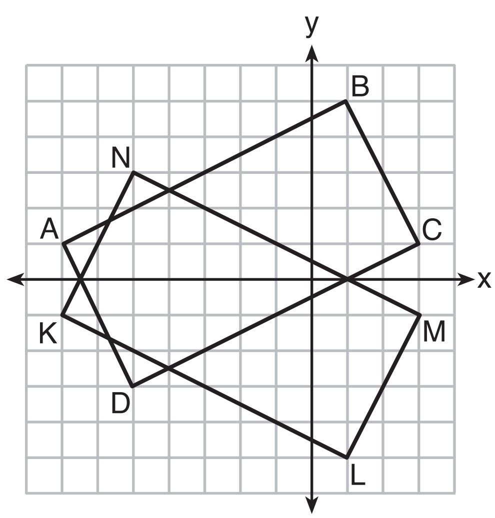
(1) $\cos J = \frac{RM}{RE}$

(2) $\cos R = \frac{JM}{JT}$

(3) $\tan T = \frac{RM}{EM}$

(4) $\tan E = \frac{TM}{JM}$

16 On the set of axes below, rectangle *ABCD* can be proven congruent to rectangle *KLMN* using which transformation?



(1) rotation

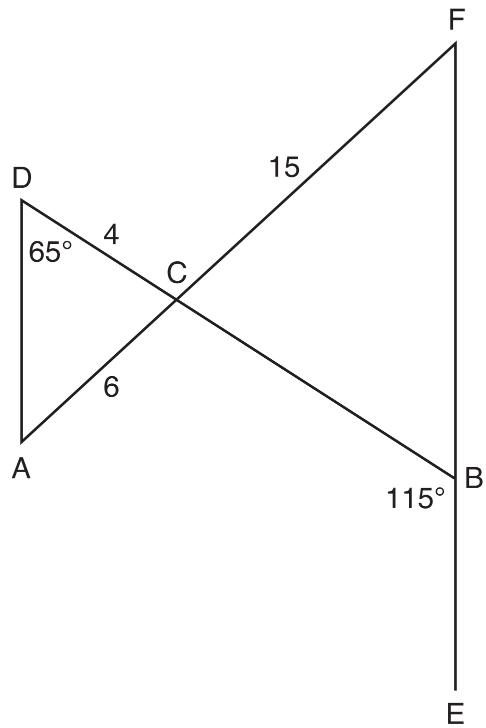
(2) translation

(3) reflection over the x-axis

(4) reflection over the y-axis

Use this space for computations.

17 In the diagram below, $\overline{DB}$ and $\overline{AF}$ intersect at point C, and $\overline{AD}$ and $\overline{FBE}$ are drawn.



If $AC = 6$, $DC = 4$, $FC = 15$, $m\angle D = 65°$, and $m\angle CBE = 115°$, what is the length of $\overline{CB}$?

(1) 10 (3) 17

(2) 12 (4) 22.5

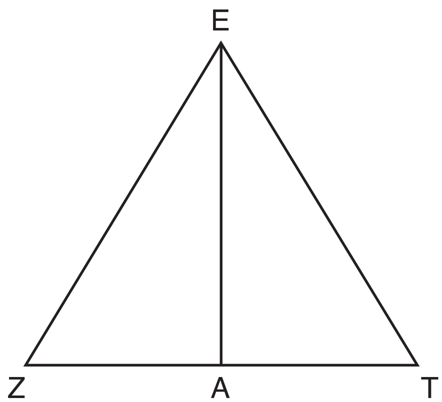
18 Seawater contains approximately 1.2 ounces of salt per liter on average. How many gallons of seawater, to the *nearest tenth of a gallon*, would contain 1 pound of salt?

(1) 3.3 (3) 4.7

(2) 3.5 (4) 13.3

Use this space for computations.

19 Line segment EA is the perpendicular bisector of $\overline{ZT}$, and $\overline{ZE}$ and $\overline{TE}$ are drawn.



Which conclusion can *not* be proven?

(1) $\overline{EA}$ bisects angle ZET.

(2) Triangle EZT is equilateral.

(3) $\overline{EA}$ is a median of triangle EZT.

(4) Angle Z is congruent to angle T.

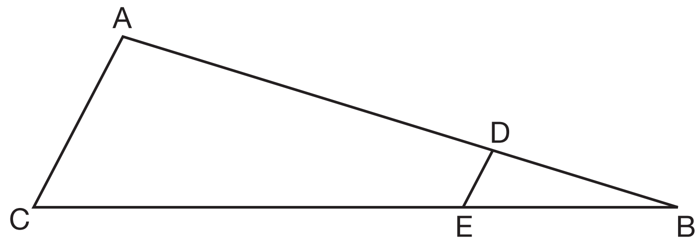
20 A hemispherical water tank has an inside diameter of 10 feet. If water has a density of 62.4 pounds per cubic foot, what is the weight of the water in a full tank, to the *nearest pound*?

(1) 16,336 (3) 130,690

(2) 32,673 (4) 261,381

Use this space for computations.

21 In the diagram of $\triangle ABC$, points D and E are on $\overline{AB}$ and $\overline{CB}$, respectively, such that $\overline{AC} \parallel \overline{DE}$.

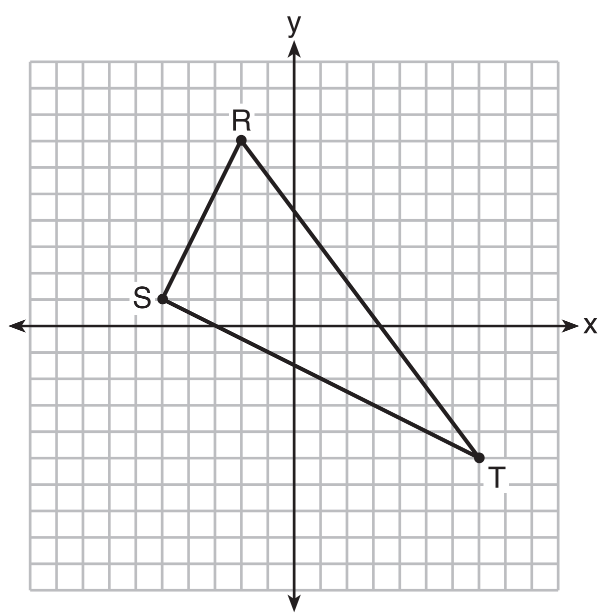


If $AD = 24$, $DB = 12$, and $DE = 4$, what is the length of $\overline{AC}$?

(1) 8 (3) 16

(2) 12 (4) 72

22 Triangle RST is graphed on the set of axes below.



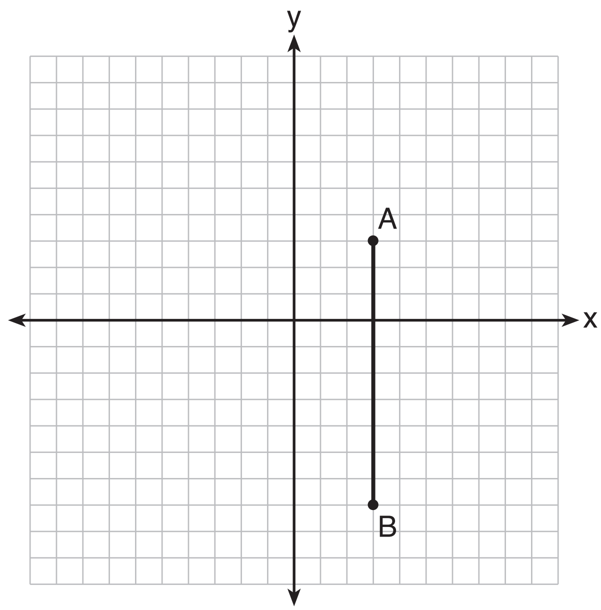
How many square units are in the area of $\triangle RST$?

(1) $9\sqrt{3} + 15$ (3) 45

(2) $9\sqrt{5} + 15$ (4) 90

Use this space for computations.

23 The graph below shows $\overline{AB}$, which is a chord of circle O. The coordinates of the endpoints of $\overline{AB}$ are $A(3,3)$ and $B(3,-7)$. The distance from the midpoint of $\overline{AB}$ to the center of circle O is 2 units.



What could be a correct equation for circle O?

(1) $(x - 1)^2 + (y + 2)^2 = 29$

(2) $(x + 5)^2 + (y - 2)^2 = 29$

(3) $(x - 1)^2 + (y - 2)^2 = 25$

(4) $(x - 5)^2 + (y + 2)^2 = 25$

24 What is the area of a sector of a circle with a radius of 8 inches and formed by a central angle that measures 60°?

(1) $\frac{8\pi}{3}$

(2) $\frac{16\pi}{3}$

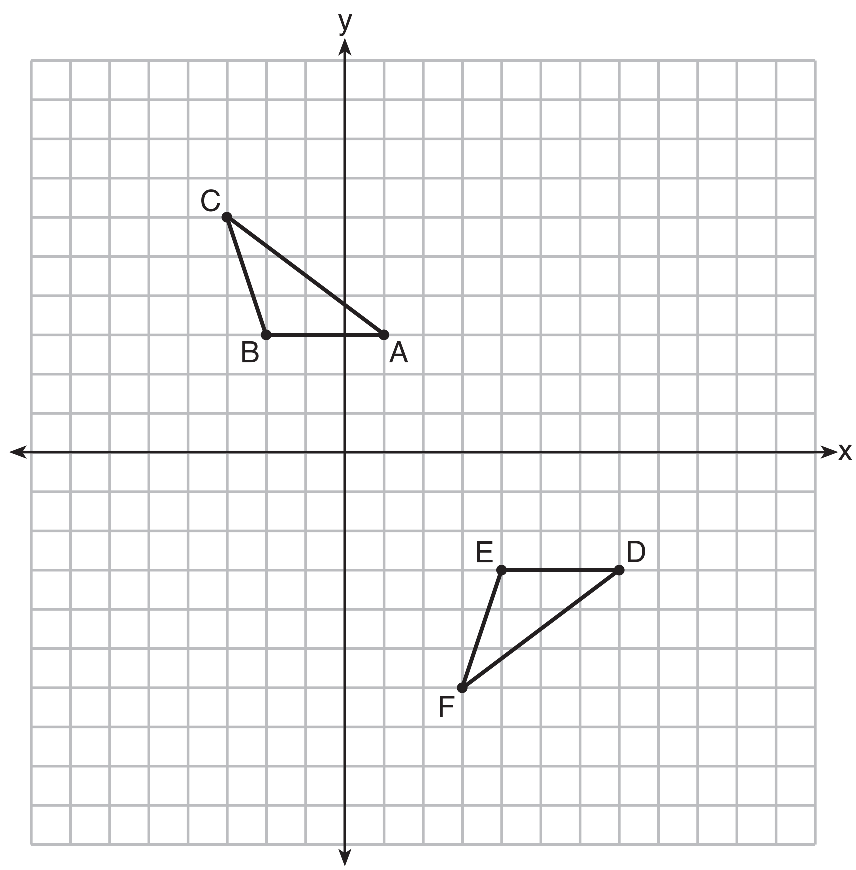
(3) $\frac{32\pi}{3}$

(4) $\frac{64\pi}{3}$

Part II

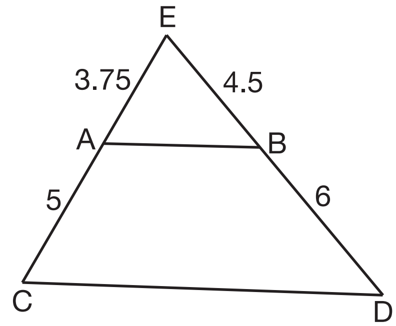
Answer all 7 questions in this part. Each correct answer will receive 2 credits. Clearly indicate the necessary steps, including appropriate formula substitutions, diagrams, graphs, charts, etc. Utilize the information provided for each question to determine your answer. Note that diagrams are not necessarily drawn to scale. For all questions in this part, a correct numerical answer with no work shown will receive only 1 credit. All answers should be written in pen, except for graphs and drawings, which should be done in pencil. [14]

25 Describe a sequence of transformations that will map $\triangle ABC$ onto $\triangle DEF$ as shown below.



26 Point P is on segment AB such that $AP:PB$ is 4:5. If A has coordinates (4,2), and B has coordinates (22,2), determine and state the coordinates of P.

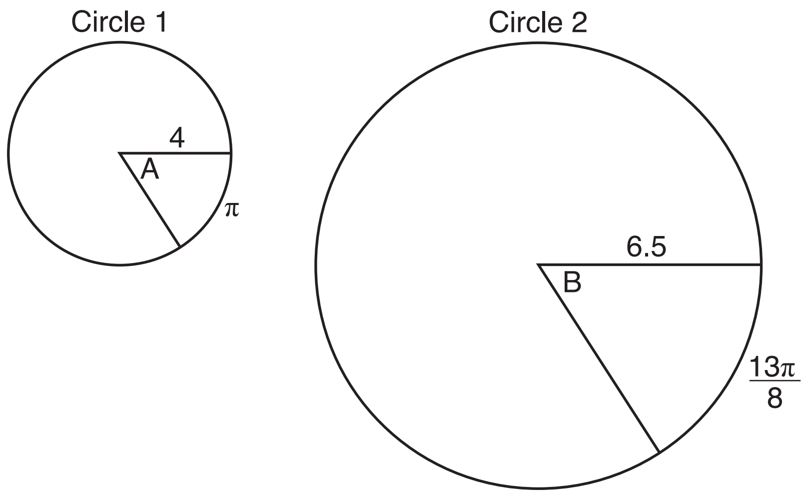
27 In $\triangle CED$ as shown below, points A and B are located on sides $\overline{CE}$ and $\overline{ED}$, respectively. Line segment AB is drawn such that $AE = 3.75$, $AC = 5$, $EB = 4.5$, and $BD = 6$.



Explain why $\overline{AB}$ is parallel to $\overline{CD}$.

28 Find the value of R that will make the equation $\sin 73° = \cos R$ true when $0° < R < 90°$. Explain your answer.

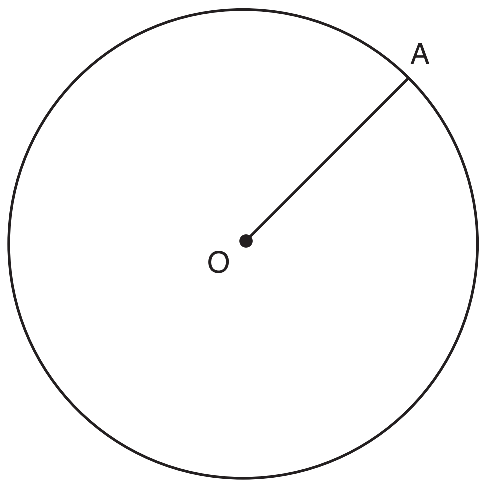
29 In the diagram below, Circle 1 has radius 4, while Circle 2 has radius 6.5. Angle A intercepts an arc of length π, and angle B intercepts an arc of length $\frac{13\pi}{8}$.



Dominic thinks that angles A and B have the same radian measure. State whether Dominic is correct or not. Explain why.

30 A ladder leans against a building. The top of the ladder touches the building 10 feet above the ground. The foot of the ladder is 4 feet from the building. Find, to the *nearest degree*, the angle that the ladder makes with the level ground.

31 In the diagram below, radius $\overline{OA}$ is drawn in circle O. Using a compass and a straightedge, construct a line tangent to circle O at point A. [Leave all construction marks.]

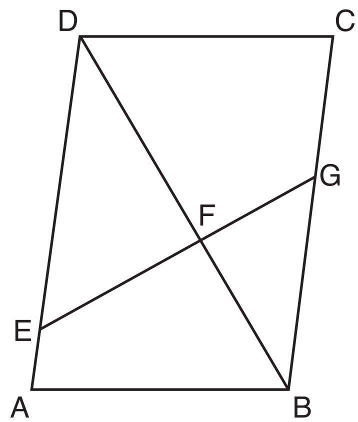


Part III

Answer all 3 questions in this part. Each correct answer will receive 4 credits. Clearly indicate the necessary steps, including appropriate formula substitutions, diagrams, graphs, charts, etc. Utilize the information provided for each question to determine your answer. Note that diagrams are not necessarily drawn to scale. For all questions in this part, a correct numerical answer with no work shown will receive only 1 credit. All answers should be written in pen, except for graphs and drawings, which should be done in pencil. [12]

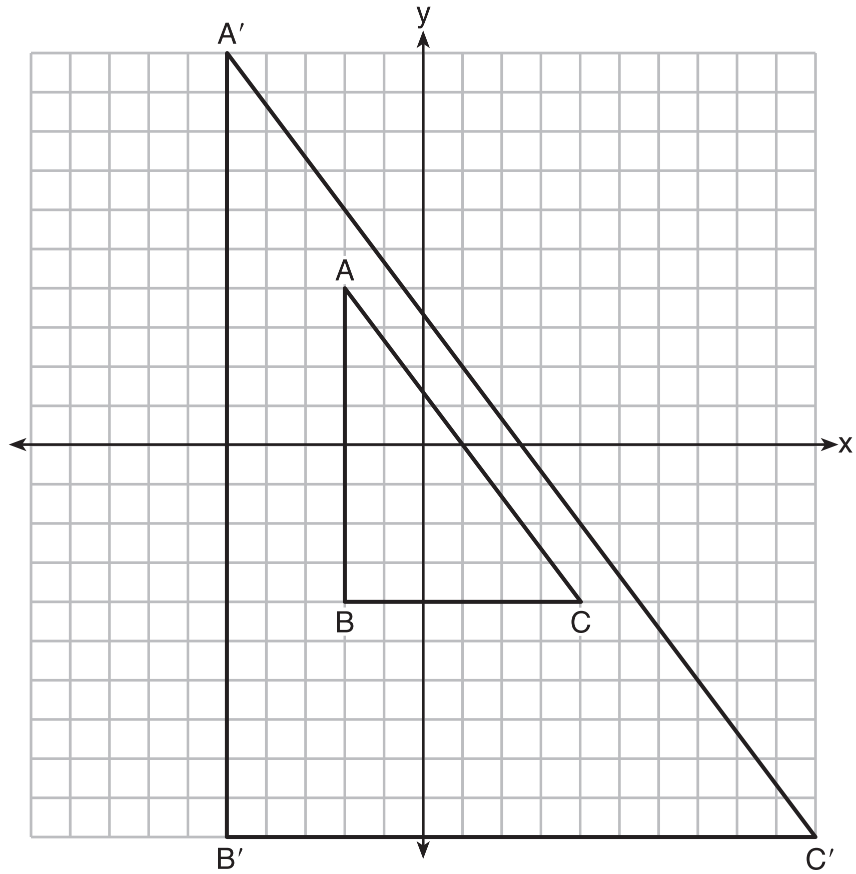
32 A barrel of fuel oil is a right circular cylinder where the inside measurements of the barrel are a diameter of 22.5 inches and a height of 33.5 inches. There are 231 cubic inches in a liquid gallon. Determine and state, to the *nearest tenth*, the gallons of fuel that are in a barrel of fuel oil.

33 Given: Parallelogram $ABCD$, $\overline{EFG}$, and diagonal $\overline{DFB}$



Prove: $\triangle DEF \sim \triangle BGF$

34 In the diagram below, $\triangle A'B'C'$ is the image of $\triangle ABC$ after a transformation.



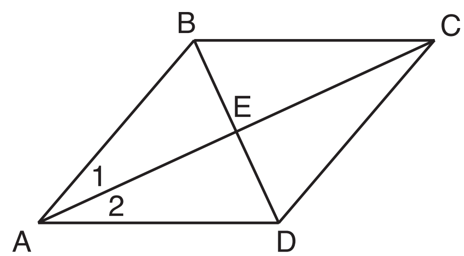
Describe the transformation that was performed.

Explain why $\triangle A'B'C' \sim \triangle ABC$.

Part IV

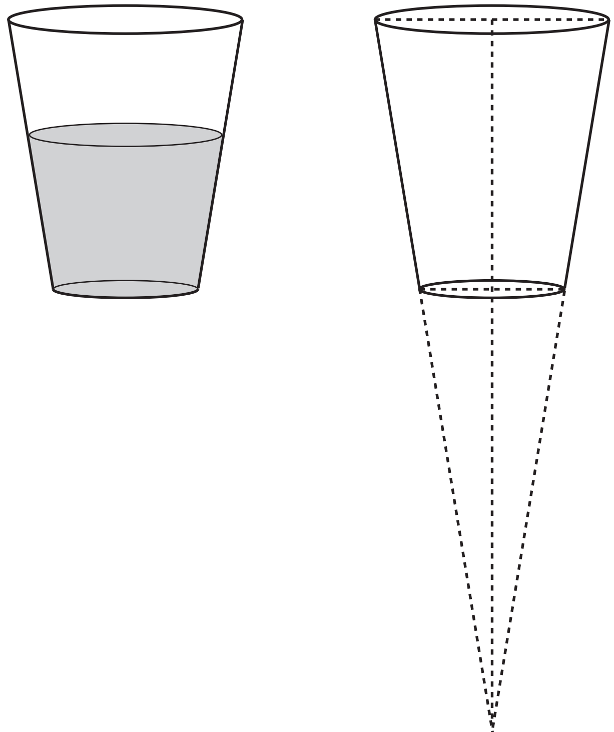
Answer the 2 questions in this part. Each correct answer will receive 6 credits. Clearly indicate the necessary steps, including appropriate formula substitutions, diagrams, graphs, charts, etc. Utilize the information provided for each question to determine your answer. Note that diagrams are not necessarily drawn to scale. For all questions in this part, a correct numerical answer with no work shown will receive only 1 credit. All answers should be written in pen, except for graphs and drawings, which should be done in pencil. [12]

35 Given: Quadrilateral *ABCD* with diagonals $\overline{AC}$ and $\overline{BD}$ that bisect each other, and $\angle 1 \cong \angle 2$



Prove: $\triangle ACD$ is an isosceles triangle and $\triangle AEB$ is a right triangle

36 A water glass can be modeled by a truncated right cone (a cone which is cut parallel to its base) as shown below.



The diameter of the top of the glass is 3 inches, the diameter at the bottom of the glass is 2 inches, and the height of the glass is 5 inches.

The base with a diameter of 2 inches must be parallel to the base with a diameter of 3 inches in order to find the height of the cone. Explain why.

Question 36 is continued on the next page.

Question 36 continued

Determine and state, in inches, the height of the larger cone.

Determine and state, to the *nearest tenth of a cubic inch*, the volume of the water glass.

If the student's responses for the multiple-choice questions are being hand scored prior to being scanned, the scorer must be careful not to make any marks on the answer sheet except to record the scores in the designated score boxes. Marks elsewhere on the answer sheet will interfere with the accuracy of the scanning.

Part I

Allow a total of 48 credits, 2 credits for each of the following. Allow credit if the student has written the correct answer instead of the numeral 1, 2, 3, or 4.

(1) 3	(9) 1	(17) . . . . . 1 . . . . .
(2) 4	(10) 2	(18) . . . . . 2 . . . . .
(3) 2	(11) 4	(19) . . . . . 2 . . . . .
(4) 1	(12) 1	(20) . . . . . 1 . . . . .
(5) 3	(13) 2	(21) . . . . . 2 . . . . .
(6) 4	(14) 4	(22) . . . . . 3 . . . . .
(7) 3	(15) 4	(23) . . . . . 1 . . . . .
(8) 4	(16) 3	(24) . . . . . 3 . . . . .

Updated information regarding the rating of this examination may be posted on the New York State Education Department's web site during the rating period. Check this web site at: http://www.p12.nysed.gov/assessment/ and select the link "Scoring Information" for any recently posted information regarding this examination. This site should be checked before the rating process for this examination begins and several times throughout the Regents Examination period.

The Department is providing supplemental scoring guidance, the "Model Response Set," for the Regents Examination in Geometry (Common Core). This guidance is intended to be part of the scorer training. Schools should use the Model Response Set along with the rubrics in the Scoring Key and Rating Guide to help guide scoring of student work. While not reflective of all scenarios, the Model Response Set illustrates how less common student responses to constructed-response questions may be scored. The Model Response Set will be available on the Department's web site at: http://www.nysedregents.org/geometrycc/.

Map to the Common Core Learning Standards
Geometry (Common Core)
June 2016

Question	Type	Credits	Cluster
1	Multiple Choice	2	G-GMD.B
2	Multiple Choice	2	G-SRT.A
3	Multiple Choice	2	G-GPE.A
4	Multiple Choice	2	G-CO.A
5	Multiple Choice	2	G-SRT.A
6	Multiple Choice	2	G-GMD.A
7	Multiple Choice	2	G-CO.C
8	Multiple Choice	2	G-CO.A
9	Multiple Choice	2	G-CO.C
10	Multiple Choice	2	G-CO.C
11	Multiple Choice	2	G-SRT.C
12	Multiple Choice	2	G-GPE.B
13	Multiple Choice	2	G-SRT.B
14	Multiple Choice	2	G-GPE.B
15	Multiple Choice	2	G-SRT.C
16	Multiple Choice	2	G-CO.B
17	Multiple Choice	2	G-CO.C
18	Multiple Choice	2	G-MG.A
19	Multiple Choice	2	G-GMD.A
20	Multiple Choice	2	G-C.A
21	Multiple Choice	2	G-SRT.B
22	Multiple Choice	2	G-GPE.B
23	Multiple Choice	2	G-GPE.A
24	Multiple Choice	2	G-C.B
25	Constructed Response	2	G-CO.A
26	Constructed Response	2	G-GPE.B
27	Constructed Response	2	G-SRT.B
28	Constructed Response	2	G-SRT.C
29	Constructed Response	2	G-C.B
30	Constructed Response	2	G-SRT.C
31	Constructed Response	2	G-CO.D
32	Constructed Response	4	G-MG.A
33	Constructed Response	4	G-SRT.B
34	Constructed Response	4	G-SRT.A
35	Constructed Response	6	G-CO.C
36	Constructed Response	6	G-MG.A

Part II

For each question, use the specific criteria to award a maximum of 2 credits. Unless otherwise specified, mathematically correct alternative solutions should be awarded appropriate credit.

(25) **[2]** A correct sequence of transformations is described.

[1] An appropriate sequence is described, but one graphing error is made.

or

[1] An appropriate sequence is described, but one conceptual error is made.

or

[1] An appropriate sequence is described, but it is incomplete.

[0] A zero response is completely incorrect, irrelevant, or incoherent or is a correct response that was obtained by an obviously incorrect procedure.

(26) **[2]** (12,2), and correct work is shown.

[1] Appropriate work is shown, but one computational error is made.

or

[1] Appropriate work is shown, but one conceptual error is made.

or

[1] Appropriate work is shown to find 12, but no further correct work is shown.

or

[1] (12,2), but no work is shown.

[0] A zero response is completely incorrect, irrelevant, or incoherent or is a correct response that was obtained by an obviously incorrect procedure.

(27) **[2]** Correct work is shown, and a correct explanation is written.

[1] Appropriate work is shown, but one computational error is made.

or

[1] Appropriate work is shown, but one conceptual error is made.

or

[1] Appropriate work is shown, but the explanation is missing or incorrect.

[0] A correct proportion is written, but no further correct work is shown.

or

[0] A zero response is completely incorrect, irrelevant, or incoherent or is a correct response that was obtained by an obviously incorrect procedure.

(28) **[2]** 17, and a correct explanation is written.

[1] Appropriate work is shown, but one computational error is made.

or

[1] Appropriate work is shown, but one conceptual error is made.

or

[1] Appropriate work is shown, but the explanation is incomplete.

or

[1] 17, but the explanation is missing or incorrect.

[0] A zero response is completely incorrect, irrelevant, or incoherent or is a correct response that was obtained by an obviously incorrect procedure.

(29) **[2]** Yes, and a correct explanation is written.

[1] Appropriate work is shown, but one computational error is made.

or

[1] Appropriate work is shown, but one conceptual error is made.

or

[1] Yes, and appropriate work is shown, but the explanation is missing or incorrect.

[0] Yes, and a correct proportion is written, but no further correct work is shown.

or

[0] A zero response is completely incorrect, irrelevant, or incoherent or is a correct response that was obtained by an obviously incorrect procedure.

(30) **[2]** 68, and appropriate work is shown.

[1] Appropriate work is shown, but one computational or rounding error is made.

or

[1] Appropriate work is shown, but one conceptual error is made.

or

[1] Tan $x = \frac{10}{4}$ or an equivalent equation is written, but no further correct work is shown.

or

[1] 68, but no work is shown.

[0] A zero response is completely incorrect, irrelevant, or incoherent or is a correct response that was obtained by an obviously incorrect procedure.

(31) **[2]** A correct construction is drawn showing all appropriate arcs.

[1] An appropriate construction is drawn showing all appropriate arcs, but the tangent line is not drawn.

[0] A drawing that is not an appropriate construction is shown.

or

[0] A zero response is completely incorrect, irrelevant, or incoherent or is a correct response that was obtained by an obviously incorrect procedure.

Part III

For each question, use the specific criteria to award a maximum of 4 credits. Unless otherwise specified, mathematically correct alternative solutions should be awarded appropriate credit.

(32) **[4]** 57.7, and correct work is shown.

[3] Appropriate work is shown, but one computational or rounding error is made.

[2] Appropriate work is shown, but two or more computational or rounding errors are made.

or

[2] Appropriate work is shown, but one conceptual error is made.

or

[2] The volume of a barrel is found in cubic inches, but no further correct work is shown.

[1] Appropriate work is shown, but one conceptual error and one computational or rounding error are made.

or

[1] 57.7, but no work is shown.

[0] A zero response is completely incorrect, irrelevant, or incoherent or is a correct response that was obtained by an obviously incorrect procedure.

(33) **[4]** A complete and correct proof that includes a concluding statement is written.

[3] A proof is written that demonstrates a thorough understanding of the method of proof and contains no conceptual errors, but one statement and/or reason is missing or incorrect or no concluding statement is written.

[2] A proof is written that demonstrates a good understanding of the method of proof and contains no conceptual errors, but two statements and/or reasons are missing or incorrect.

or

[2] A proof is written that demonstrates a good understanding of the method of proof, but one conceptual error is made.

[1] Only one correct statement and reason are written.

[0] The "given" and/or the "prove" statements are written, but no further correct relevant statements are written.

or

[0] A zero response is completely incorrect, irrelevant, or incoherent or is a correct response that was obtained by an obviously incorrect procedure.

(34) **[4]** Dilation of $\frac{5}{2}$ centered at the origin is written. A correct explanation is written.

[3] Appropriate work is shown, but one computational error is made.

or

[3] The description of the dilation is incomplete. A correct explanation is written.

or

[3] A correct description of a dilation is written. An appropriate explanation is written, but it is incomplete.

[2] A correct description of a dilation is written, but the explanation is missing.

or

[2] The description of the dilation is incomplete. An appropriate explanation is written, but it is incomplete.

or

[2] A correct explanation is written, but no further correct work is shown.

[1] The description of the dilation is incomplete. No further correct work is shown.

or

[1] An appropriate explanation is written, but it is incomplete. No further correct work is shown.

[0] A zero response is completely incorrect, irrelevant, or incoherent or is a correct response that was obtained by an obviously incorrect procedure.

Part IV

For each question, use the specific criteria to award a maximum of 6 credits. Unless otherwise specified, mathematically correct alternative solutions should be awarded appropriate credit.

(35) **[6]** A complete and correct proof is written that includes both concluding statements.

[5] A proof is written that demonstrates a thorough understanding of the method of proof and contains no conceptual errors, but one statement and/or reason is missing or incorrect.

[4] A proof is written that demonstrates a thorough understanding of the method of proof and contains no conceptual errors, but two statements and/or reasons are missing or incorrect.

or

[4] $\triangle ACD$ is an isosceles triangle or $\triangle AEB$ is a right triangle is proven, but no further correct work is shown.

[3] A proof is written that demonstrates a thorough understanding of the method of proof, but three statements and/or reasons are incorrect.

or

[3] A proof is written that demonstrates a thorough understanding of the method of proof, but one conceptual error is made.

[2] A proof is written that demonstrates a thorough understanding of the method of proof, but one conceptual error is made, and one statement and/or reason is missing or incorrect.

or

[2] Some correct relevant statements about the proof are made, but four statements and/or reasons are missing or incorrect.

or

[2] Appropriate work is shown to prove *ABCD* is a rhombus, but no further correct work is shown.

[1] Only one or two correct relevant statements and reasons are written.

[0] The "given" and/or the "prove" statements are written, but no further correct relevant statements are written.

or

[0] A zero response is completely incorrect, irrelevant, or incoherent or is a correct response that was obtained by an obviously incorrect procedure.

(36) **[6]** 15, 24.9, and correct work is shown. A correct explanation is written.

[5] Appropriate work is shown, but one computational or rounding error is made.

or

[5] 15, 24.9, and correct work is shown, but the explanation is missing or incorrect.

[4] Appropriate work is shown, but two computational or rounding errors are made.

or

[4] Appropriate work is shown, but one conceptual error is made in finding the volume of the glass.

[3] Appropriate work is shown, but three or more computational or rounding errors are made.

or

[3] Appropriate work is shown, but one conceptual error in finding the volume of the glass and one computational or rounding error are made.

[2] Appropriate work is shown, but one conceptual error in finding the volume of the glass and two computational or rounding errors are made.

or

[2] Correct work is shown to find 15, the height of the cone, but no further correct work is shown.

[1] A correct explanation is written, but no further correct work is shown.

or

[1] Correct work is shown to find the volume of either cone, but no further correct work is shown.

or

[1] 15, and 24.9, but no work is shown.

[0] A zero response is completely incorrect, irrelevant, or incoherent or is a correct response that was obtained by an obviously incorrect procedure.